

Project Export

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1. Market Basket Analysis for Upselling/Cross-Selling

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Summary

This project involves applying association rule mining to identify frequently co-occurring product purchases, enabling targeted upselling and cross-selling strategies. Certification signifies a strong understanding of pattern recognition, association rules, and their application in retail analytics.

Full Content

Project Overview

Have you ever noticed that when you buy a loaf of bread, the grocery store places the butter and jam nearby? Or when you buy a smartphone online, the site suggests a screen protector? This is not intuition; it is Market Basket Analysis (MBA).

As a Data Science Intern focusing on Retail Analytics, your goal is to uncover hidden patterns in transactional data. You will use Association Rule Mining to discover relationships between products that are frequently purchased together. The insights you generate will directly influence store layout design, website recommendation widgets, and promotional bundle offers, ultimately driving the Average Order Value (AOV) up.

Key Objectives & Workflow

1. Data Preprocessing & Transformation

- The Challenge: Transactional data usually comes in a "Long" format (e.g., TransactionID: 101, Item: Milk on one row, TransactionID: 101, Item: Bread on the next). Most algorithms require a "Basket" format.
- The Task: You must write efficient code to transform this raw data. You will group items by Transaction ID and convert them into either a List of Lists or a One-Hot Encoded Matrix (where columns are products and rows are transactions, marked 1 or 0).

2. Association Rule Mining (The Algorithms)

- Apriori Algorithm: Implement the classic Apriori algorithm. You will iteratively find "frequent itemsets" based on a minimum support threshold.
- FP-Growth (Frequent Pattern Growth): Since Apriori can be computationally expensive on large datasets, you will also implement the FP-Growth algorithm (which uses a tree structure) and compare the execution time between the two.

3. Metric Calculation & Filtering

- Understanding the Math: You cannot just output every rule. You must filter them based on

three critical metrics:

- Support: How popular is the itemset? ($\text{\$Frequency}(A,B) / \text{Total Transactions}\$$)
- Confidence: Likelihood of buying B if A is bought. ($\text{\$Frequency}(A,B) / \text{Frequency}(A)\$$)
- Lift: The ratio of the observed support to that expected if A and B were independent. ($\text{\$Confidence} / \text{Support}(B)\$$). Note: A Lift > 1 indicates a strong association.
- Filtering Logic: You will create a script that filters specifically for High Lift rules, as these represent true "hidden" connections rather than just popular items.

4. Network Visualization

- Numbers in a table can be hard to interpret. You will use Graph Theory libraries to plot the product relationships as a network node graph, where node size represents product popularity and edge thickness represents the strength of the association (Lift).

Prerequisites

To successfully complete this project, you should possess the following skills:

- Programming: Python.
- Libraries: Pandas, MLxtend (the industry standard for MBA), NetworkX (for graph visualization), Matplotlib.
- Concepts: Probability, Set Theory, Association Rules (Apriori, Eclat, FP-Growth), Data Transformation.

Deliverables

1. The Rules Engine (Python Script)

- A script that ingests a raw CSV of transactions and outputs a clean list of Association Rules sorted by Lift.

2. The Visual Network

- A high-resolution image or interactive HTML file (using PyVis or NetworkX) showing the constellation of products. This helps stakeholders visually spot clusters (e.g., the "Breakfast Cluster" or the "Cleaning Supplies Cluster").

3. Business Strategy Proposal

- A PDF document translating the math into English. It should contain concrete recommendations, such as:
 - "Create a 'Weekend Bundle' containing Product X and Product Y."
 - "Move Product A closer to the checkout counter when Product B is on sale."

Project Timeline & Deadlines

- Week 1: Data Formatting & EDA
 - Goal: Prepare the "Basket".
 - Task: Clean the data (remove cancelled orders). Convert the dataframe from 'Long' to 'One-Hot Encoded' format. Analyze top 10 best-selling items.
- Week 2: Generating Itemsets (Apriori/FP-Growth)
 - Goal: Find frequent patterns.
 - Task: Use mlxtend to generate frequent itemsets. Experiment with different min_support values (e.g., 0.01 vs 0.05) to see how the number of results changes.
- Week 3: Rule Generation & Metric Analysis
 - Goal: Derive logic from patterns.
 - Task: Calculate Confidence and Lift for the itemsets. Prune the rules (e.g., keep only rules with Confidence > 0.6 and Lift > 1.2).
- Week 4: Visualization & Strategy
 - Goal: Business impact.
 - Task: Build the Network Graph. Interpret the strongest rules and write the strategy proposal.